Probability and Stochastic Processes
Coursework

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1. a) i) $E(Y) = E(2X+4) = 2E(X)+4 = 14$

ii) $D^2X = E(X^2) - E^2(X) \Rightarrow E(X^2) = 29$, $E(Y^2) = E[(2X+4)^2] = 212$

$D^2Y = E(Y^2) - E^2(Y) \Rightarrow D^2Y = 16 \Rightarrow DY = 4$

iii) $f_X(x) = \frac{1}{\sqrt{2\pi\sigma^2}} e^{-(x-\mu)^2/2\sigma^2}$ where $\mu = 5$, $\sigma = 2$

$F_Y(y) = P(Y \le y) = P(2X+4 \le y) = P(X \le \frac{y-4}{2}) = F_X(\frac{y-4}{2})$

$f_Y(y) = F_Y(y)' = \frac{1}{2} f_X(\frac{y-4}{2})$

hence $f_Y(y) = \frac{1}{4\sqrt{2\pi}} e^{-\frac{(y-14)^2}{32}}$ ✓

b)

$f_X(x) = \begin{cases} 1 & , 0 \le x \le 1 \\ 0 & , \text{otherwise} \end{cases}$

$F_Y(y) = P(Y \le y) = P(-\ln x \le y)$; $x = h(y) = e^{-y}$

$f_Y(y) = F_Y'(y) = \begin{cases} e^{-y} & , 0 < y < \infty \\ 0 & , \text{otherwise} \end{cases}$ ✓

c) i) $f_Z(z) = f_{X+Y}(z) = \int_{-\infty}^{\infty} f_X(z-y) f_Y(y) dy$

$= \int_0^z e^{-z} dy = z e^{-z}$, $z > 0$ ✓

ii) $F_{min}(z) = 1 - [1 - F_X(z)][1 - F_Y(z)]$

$F_X(x) = 1 - e^{-x}$, $x > 0$ $\quad F_Y(y) = 1 - e^{-y}$, $y > 0$

$F_{min}(z) = 1 - e^{-2z}$, $z > 0$

$f_{min}(z) = 2e^{-2z}$, $z > 0$ ✓

iii) $F_{max}(z) = F_X(z) F_Y(z) = (1 - e^{-z})(1 - e^{-z})$, $z > 0$

$f_{max}(z) = -2e^{-2z} + 2e^{-z}$, $z > 0$ ✓

2. a) For n sample,

$f(\underline{x}, c) = c^{4n} (x_1 \cdots x_n)^{3n} e^{-c(x_1 + \cdots + x_n)}$

$\frac{\partial f(\underline{x}, c)}{\partial c} = 4n c^{4n-1} (x_1 \cdots x_n)^{3n} e^{-c(x_1 + \cdots + x_n)}$

$- (x_1 + \cdots + x_n) c^{4n} (x_1 \cdots x_n)^{3n} e^{-c(x_1 + \cdots + x_n)}$

$= \frac{4n f(\underline{x}, c)}{c} - (x_1 + \cdots + x_n) f(\underline{x}, c)$

$= [\frac{4n}{c} - (x_1 + \cdots + x_n)] f(\underline{x}, c)$

$\frac{\partial f(\underline{x}, c)}{\partial c} = 0 \Rightarrow c = \frac{4n}{x_1 + \cdots + x_n} = \frac{4 \times 3}{3.1 + 3.4 + 3.3} = \frac{60}{49}$ ✓

b) $f_X(x) = \begin{cases} 1 & , 0 \le x \le 1 \\ 0 & , \text{otherwise} \end{cases}$

$F_X(x) = \begin{cases} x & , 0 \le x \le 1 \\ 0 & , \text{otherwise} \end{cases}$

$F_{max}(Y) = \begin{cases} F_{X_1}(y) \cdots F_{X_i}(y) & , i = n, 0 \le y \le 1 \\ 0 & , \text{otherwise} \end{cases}$

$= \begin{cases} y^n & , 0 \le y \le 1 \\ 0 & , \text{otherwise} \end{cases}$ ✓

c) MMSE

$$\begin{cases} E\{[s(t-T/2)-as(t)-bs(t-T)]\,s^*(t)\}=0 & ① \\ E\{[s(t-T/2)-as(t)-bs(t-T)]\,s^*(t-T)\}=0 & ② \end{cases}$$

① $E\{s(t-T/2)s^*(t)-as(t)s^*(t)-bs(t-T)s^*(t)\}=0$

$$R_s(T/2)-aR_s(0)-bR_s(T)=0$$

$$R_s(T/2)-bR_s(T)=aR_s(0)$$

② $E\{s(t-T/2)s^*(t-T)-as(t)s^*(t-T)-bs(t-T)s^2(t-T)\}=0$

$$R_s(T/2)-aR_s(T)=bR_s(0) \quad , R_s(\tau)=1\cdot e^{-|\tau|/T}$$

①, ② $\Rightarrow a=b=\frac{\sqrt{e}}{1+e}$ ✓

$E\{[s(t-T/2)-as(t)-bs(t-T)]^2\}=E[s^2(t-T/2)-as(t-T/2)s^*(t)$

$-bs(t-T/2)s^*(t-T)-as(t)s^*(t-T/2)+a^2s^2(t)+abs(t)s^*(t-T)$

$-bs(t-T/2)s^*(t-T)+abs(t)s^*(t-T)+b^2s^2(t-T)]$

$=E[-2as(t-T/2)s^*(t)+2abs(t)s^*(t-T)-2bs(t-T/2)s^*(t-T)]$

$+(1+a^2+b^2)R_s(0)$

$=-2aR_s(T/2)+2abR_s(T)-2bR_s(T/2)+(1+a^2+b^2)R_s(0)$

$=2a^2R_s(T)-4aR_s(T/2)+(1+2a^2)R_s(0)$

$=\frac{(e+1)(e-1)2}{(-e+1)^2}=\frac{e-1}{e+1}$ ✓

3. a) i) $P(\text{head})=\frac{1}{2}$, $P(\text{tail})=\frac{1}{2}$

$E[X(t)]=\frac{1}{2}\sin\pi t+\frac{1}{2}\cdot 2t=\frac{1}{2}\sin\pi t+t$ ✓

ii) $X(\frac{1}{4})=\begin{cases}\frac{\sqrt{2}}{2}, & \text{head}\\ \frac{1}{2}, & \text{tail}\end{cases}$

$X(\frac{1}{4})$	$\frac{\sqrt{2}}{2}$	$\frac{1}{2}$
P^k	$\frac{1}{2}$	$\frac{1}{2}$

$$F(x,\tfrac{1}{4})=\begin{cases}0 & x<\frac{1}{2}\\ \frac{1}{2} & \frac{1}{2}\le x<\frac{\sqrt{2}}{2}\\ 1 & x\ge\frac{\sqrt{2}}{2}\end{cases}$$ ✓

$X(\frac{1}{2})=\begin{cases}1, & \text{head}\\ 1, & \text{tail}\end{cases}$

$X(\frac{1}{2})$	1
P^k	1

$$F(x,\tfrac{1}{2})=\begin{cases}0 & x<1\\ 1 & x\ge 1\end{cases}$$ ✓

$X(1)=\begin{cases}0, & \text{head}\\ 2, & \text{tail}\end{cases}$

$X(1)$	0	2
P^k	$\frac{1}{2}$	$\frac{1}{2}$

$$F(x,1)=\begin{cases}0 & x<0\\ \frac{1}{2} & 0\le x<2\\ 1 & x\ge 2\end{cases}$$ ✓

b) i) $P\{|X(t+\tau)-X(t)|\ge a\}\le E[|X(t+\tau)-X(t)|]/a$

$\Rightarrow P\{|X(t+\tau)-X(t)|^2\ge a\}\le E[|X(t+\tau)-X(t)|^2]/a^2$

$=E[X(t+\tau)^2-2X(t+\tau)X(t)+X(t)^2]/a^2$

$=\frac{E[X(t+\tau)^2-2E[X(t+\tau)X(t)]+E[X(t)^2]}{a^2}$

$=\frac{-2E[X(t+\tau)X(t)]}{a^2}=\frac{2R(0)-2R(1)}{a^2}$

hence $P\{|X(t+\tau)-X(t)|\ge a\}\le\frac{2R(0)-2R(1)}{a^2}$ ✓

i) $f(x_1, x_2; \tau) = f(x_1, x_2; t_1 - t_2) \Rightarrow \tau = t_1 - t_2 \Rightarrow t_1 = t_2 + \tau$

$P\{|X(t+\tau) - X(t)| \geq a\} = P\{|X(t_1) - X(t_2)| \geq a\}$

$= 1 - \int_{-\infty}^{+\infty} \int_{x_1(t)-a}^{x_1(t)+a} f(x_1, x_2; \tau) dx_2 dx_1$

c) $E\{X(t)\} = \mu$, $E\{X(t_1) X^*(t_2)\} = R_{XX}(t_1 - t_2) = 5\delta(t)$

$Y'(t) + 2Y(t) = X(t) \Rightarrow jwY(w) + 2Y(w) = X(w)$

$\Rightarrow H(w) = \frac{1}{2 + jw}$

$R_X(\tau) = 5\delta(\tau) \Rightarrow R_X(w) = 5$

$R_{XY}(w) = R_{XX}(w) \cdot H(w) = \frac{5}{2 + jw}$

$R_{XY}(t_1, t_2) = 5e^{-2t} u(t)$ ~~t~~ τ (-1)

$R_{YY}(w) = R_{XY}(w) \cdot H^*(w) = \frac{5}{4 + w^2}$

$R_{YY}(t_1, t_2) = \frac{5}{4} e^{-2t} u(t)$ (-2)

$E[Y^2(t)] = R_Y(0) = \frac{5}{4}$ ✓

d) $H(w) = \frac{1}{-w^2 + 2jw + 5}$

$R_X(\tau) = \frac{1}{2\pi} \int_{-\infty}^{\infty} S_X(w) e^{jw\tau} dw = E[X^2(t)]$

$R_{YY} = R_{XX} * h(-t) * h(t)$

$R_Y(w) = \frac{1}{2\pi} \int_{-\infty}^{\infty} S_X(w) \cdot H^*(w) \cdot H(w) dw$

$= \frac{1}{2\pi} \int_{-\infty}^{\infty} \frac{S_X(w)}{(-w^2 - 2jw + 5)(w^2 + 2jw + 5)} dw = \frac{1}{2\pi} \int_{-\infty}^{\infty} \frac{S_X(w)}{(5 - w^2)^2 + 4w^2} dw$

$= \frac{1}{2\pi} \int_{-\infty}^{\infty} \frac{S_X(w)}{w^4 - 10w^2 + 25 + 4w^2} dw = \frac{1}{2\pi} \int_{-\infty}^{\infty} \frac{S_X(w)}{w^4 - 6w^2 + 25} dw$

Let maximize $R_Y(w)$

minimize $(w^4 - 6w^2 + 25) \Rightarrow w^2 = 3 \Rightarrow |w| = \sqrt{3}$

$E[X^2(t)] = R_X(0) = \frac{1}{2\pi} \int_{-\infty}^{\infty} S_X(w) dw = 10 \Rightarrow R_X(t) = 10\cos(\sqrt{3}t)$

hence, $S_X(w) = 10\pi [\delta(w - \sqrt{3}) + \delta(w + \sqrt{3})]$ ✓

4. a) Assume $t_1 < t_2 < t_3$

$E[(X(t_1)-\lambda t_1)(X(t_2)-\lambda t_2)(X(t_3)-\lambda t_3)]$

$= E[X(t_1)X(t_2)X(t_3) - \lambda t_3 X(t_1)X(t_3) - \lambda t_1 X(t_2)X(t_3) + \lambda^2 t_1 t_2 X(t_3) - \lambda t_3 X(t_1)X(t_2)$

$+ \lambda^2 t_2 t_3 X(t_1) + \lambda^2 t_1 t_3 X(t_2) - \lambda^3 t_1 t_2 t_3]$

$E[X] = \sum n \cdot \frac{\lambda^n}{n!} e^{-\lambda} = \lambda \sum \frac{\lambda^{n-1}}{(n-1)!} e^{-\lambda} = \lambda$

$E[X(t_1)X(t_2)]$, $t_1 < t_2$

$E[X(t_1)X(t_2)] = E\{X(t_1)[n(0,t_1) + n(t_1,t_2)]\}$

$= E\{n^2(0,t_1) + n(0,t_1)n(t_1,t_2)\} = E[X^2(t_1)] + E[n(0,t_1)n(t_1,t_2)]$

$E[X^2] = \sum n^2 \cdot \frac{\lambda^n}{n!} e^{-\lambda} = \sum (n-1+1) \frac{\lambda^n}{(n-1)!} e^{-\lambda} = \sum \frac{\lambda^n}{(n-2)!} e^{-\lambda} + \sum \frac{\lambda^n}{(n-1)!} e^{-\lambda}$

$= \lambda^2 + \lambda$

$E[X(t)] = \lambda t$, $E[X^2(t)] = (\lambda t)^2 + \lambda t$

$E[X(t_1)X(t_2)] = (\lambda t_1)^2 + \lambda t_1 + \lambda t_1 \cdot \lambda(t_2 - t_1) = \lambda^2 t_1 t_2 + \lambda t_1$, $t_1 < t_2$

hence, $E[X(t_1)X(t_2)] = \lambda^2 t_1 t_2 + \lambda \min(t_1, t_2)$

$E[\tilde{X}(t_1)\tilde{X}(t_2)\tilde{X}(t_3)] = E[X(t_1)X(t_2)X(t_3)] - \lambda t_2[\lambda^2 t_1 t_3 + \lambda \min(t_1, t_3)]$

$- \lambda t_1[\lambda^2 t_2 t_3 + \lambda \min(t_2, t_3)] + \lambda^3 t_1 t_2 t_3 - \lambda t_3[\lambda^2 t_1 t_2 + \lambda \min(t_1, t_2)]$

$+ \lambda^3 t_1 t_2 t_3 + \lambda^2 t_1 t_2 t_3 - \lambda^3 t_1 t_2 t_3$

$= E[X(t_1)X(t_2)X(t_3)] - \lambda^3 t_1 t_2 t_3 - \lambda^2 t_1 \min(t_2, t_3) - \lambda^2 t_2 \min(t_1, t_3)$

$- \lambda^2 t_3 \min(t_1, t_2)$

$E[X(t_1)X(t_2)X(t_3)] = E\{n(0,t_1)[n(0,t_1) + n(t_1,t_2)][n(0,t_1) + n(t_1,t_2)$

$+ n(t_2,t_2)]\}$

$= E\{n^3(0,t_1) + n^2(0,t_1)n(t_1,t_2) + n^2(0,t_1)n(t_2,t_3) + n^2(0,t_1)n(t_1,t_2)$

$+ n(0,t_1)n^2(t_1,t_2) + n(0,t_1)n(t_1,t_2)n(t_2,t_3)\}$

$E(X^3) = \sum n^3 \frac{\lambda^n}{n!} e^{-\lambda} = \sum [(n-1)(n-2) + 3n - 2] \frac{\lambda^n}{(n-1)!} e^{-\lambda}$

$= \sum [(n-1)(n-2) + 3(n-1) + 1] \frac{\lambda^n}{(n-1)!} e^{-\lambda} = \lambda^3 + 3\lambda^2 + \lambda$

hence $E[X^3(t)] = (\lambda t)^3 + 3(\lambda t)^2 + \lambda t$

$E[n^2(t_1,t_2)] = \lambda^2(t_2 - t_1)^2 + \lambda(t_2 - t_1)$, $t_1 < t_2$

$E[X(t_1)X(t_2)X(t_3)] = (\lambda t_1)^3 + 3(\lambda t_1)^2 + \lambda t_1 t_2[\lambda^2 t_1^2 + \lambda t_1)\lambda(t_2 - t_1)$

$+ (\lambda^2 t_1^2 + \lambda t_1)\lambda(t_3 - t_2) + \lambda t_1[\lambda^2(t_1 - t_2)^2 + \lambda(t_2 - t_1)]$

$+ \lambda t_1 \lambda(t_2 - t_1)\lambda(t_3 - t_2)$

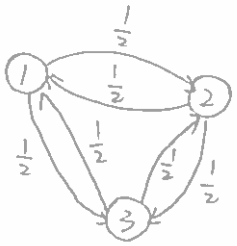
$= \lambda t_1 + \lambda^2 t_1 t_2 + \lambda^2 t_1 t_3 + \lambda^2 t_1 t_2 + \lambda^3 t_1 t_2 t_3$

hence $E[\tilde{X}(t_1)\tilde{X}(t_2)\tilde{X}(t_3)] = \lambda t_1 + 2\lambda^2 t_1 t_2 + \lambda^2 t_1 t_3 + \lambda^3 t_1 t_2 t_3$

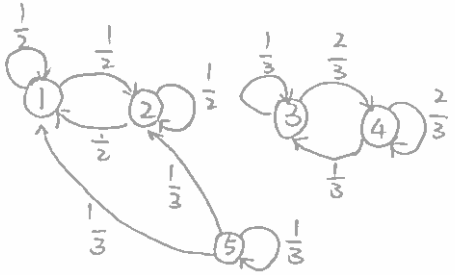
$- \lambda^3 t_1 t_2 t_3 - \lambda^2 t_1 t_2 - \lambda^2 t_2 t_1 - \lambda^2 t_1 t_3 = \lambda t_1$

$E[\tilde{X}(t_1)\tilde{X}(t_2)\tilde{X}(t_3)] = \lambda \min(t_1, t_2, t_3)$ ✓

2)

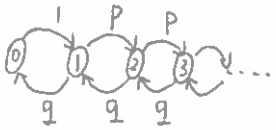


classify states? (2)



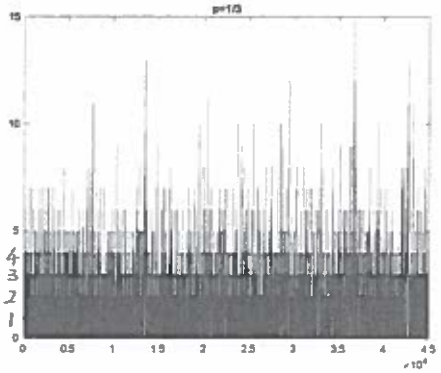
classify states? (4)

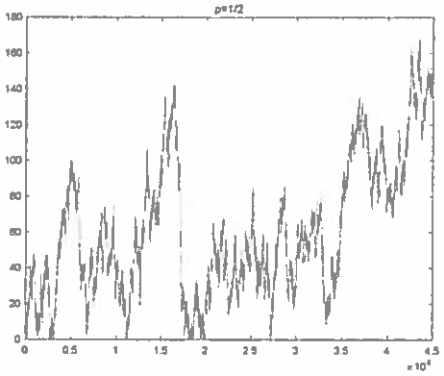
(c)

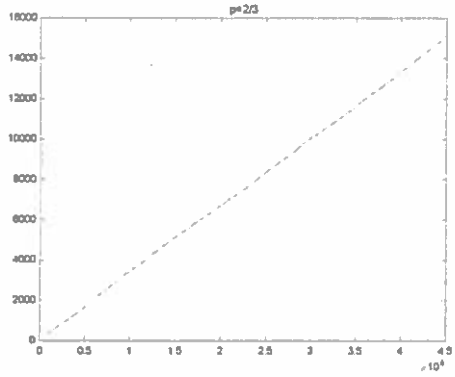


$P+q=1$

X is state







From figures above, when $P=\frac{1}{3}$, $q=\frac{2}{3}$, the probility that the man walk to left is much larger than walk to right, then the final state is small (mostly less than 5); when $P=\frac{1}{2}$, $q=\frac{1}{2}$, the probility that the man walk to right and left are the same, then the final state changes every time when I run the matlab, but the final state is less than 180 as then man will walk to both sides; when $P=\frac{2}{3}$, $q=\frac{1}{3}$, the probility that the man walk to right is much larger than walk to left, then the final state will increase with the increase of steps.

classify the states? (-3)